

→ Chapter "4"

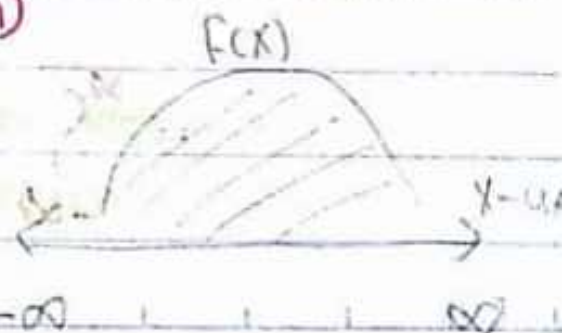
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Ex

- ① the temperature between 30 and 40 $\Rightarrow x \in [30, 40]$
- ② the time you wait for bus.

Note

② $P(X=0) = \int_c f(x) \cdot dx = \text{zero}$

$$\boxed{2} \int_{-\infty}^{\infty} f(x) \cdot dx = 1$$


(2)

Ex let p.d.f of random variable x is

$$f(x) = \begin{cases} \frac{1}{10} & : 0 \leq x \leq 10 \\ 0 & \text{o.w otherwise} \end{cases}$$

0 = x-axis (← 25)

Find:

① $P(1 \leq x \leq 3) = \int_1^3 \frac{1}{10} dx = \frac{2}{10} = \frac{1}{5}$

② $P(\frac{19}{10} \leq x \leq 2) = \int_{1.9}^2 \frac{1}{10} dx = \frac{1}{10} (2 - \frac{19}{10}) = \frac{1}{100}$

③ $P(X=4) = \text{Zero}$

Fact : If x is a continuous Random Variable whose p.d.f then:

① the expected value of x is $E(x) = \mu$

$$= \int_{-\infty}^{\infty} x \cdot f(x) \cdot dx$$

② The variance of x is $\text{Var}(x) = \sigma^2 = E(x - \mu)^2$

$$= \int_{-\infty}^{\infty} (x - \mu)^2 \cdot f(x) \cdot dx$$

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or $E(g(x)) = \int_{-\infty}^{\infty} g(x) \cdot f(x) \cdot dx$

$g(x) = x^2 \Rightarrow E(x^2) = \int_{-\infty}^{\infty} x^2 \cdot f(x) \cdot dx$

Ex Given the following P.d.F

$$f(x) = \begin{cases} cx^2 & : 0 \leq x < 1 \\ 0 & : \text{o.w} \end{cases}$$

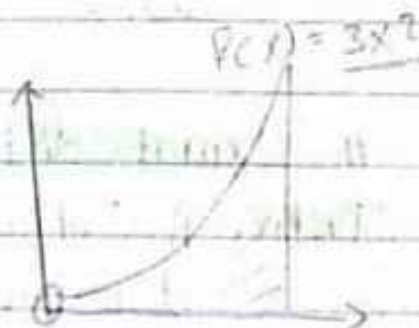
Find

- ① The value of c
- ② $P(0 \leq x < 1/2)$
- ③ $E(x)$?
- ④ σ^2 ?

Soln : II $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\int_0^1 cx^2 dx = 1$$

$$\frac{cx^3}{3} \Big|_0^1 = 1 \Rightarrow \frac{c}{3} = 1 \Rightarrow \boxed{c=3}$$



$$\text{② } P(0 \leq x < 1/2) = \int_0^{1/2} 3x^2 \cdot dx$$

$$= x^3 \Big|_0^{1/2} = \frac{1}{8}$$

$$f(x) = 3x^2$$

③

$$E(X) = \int_{-\infty}^{\infty} x \cdot P(x) \cdot dx = \int_0^1 x \cdot (3x^2) \cdot dx = \int_0^1 3x^3 \cdot dx$$

$$= \frac{3x^4}{4} \Big|_0^1 = \frac{3}{4}$$

④

$$\sigma^2 = E(X^2) - (E(X))^2 \quad E(X) = \mu = 3/4$$

$$E(X^2) = \int_0^1 x^2 \cdot f(x) \cdot dx = \int_0^1 3x^4 \cdot dx = \frac{3}{5} x^5 \Big|_0^1 = \frac{3}{5}$$

$$\sigma^2 = \frac{3}{5} - \left(\frac{3}{4}\right)^2 = 3.075$$

$$\sigma = \sqrt{\sigma^2} = \sqrt{0.0375}$$

Standard

Normal distribution

Defn: A Continuous random variable X follow a normal distribution IF it has p.d.f

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{1}{2\sigma^2} (x-\mu)^2} \quad -\infty < x < \infty$$

$$\mu = E(X) \rightarrow -\infty < \mu < \infty$$

$$\sigma^2 = \text{var}(X) \rightarrow \sigma > 0$$

Note

1. the parameter of the distribution are μ & σ^2

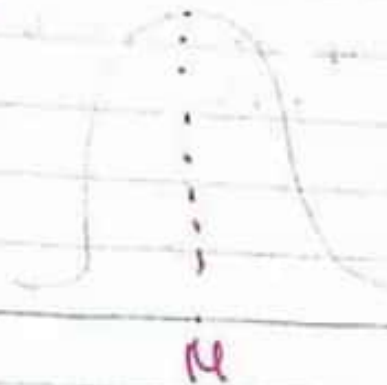
$$X \sim N(\mu, \sigma^2)$$

$$X \sim N(\mu, \sigma^2)$$

توزيع

[2.]

$$\frac{\sigma^2}{2\sigma^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



[1] Symmetric about $x = \mu$

[2] Bell-shaped

[3] Unimodal

[4] Mean = Median = Mode

[5] $\int_{-\infty}^{\infty} f(x) \cdot dx = 1$
Area under curve

Fact: IF $X \sim N(\mu, \sigma^2)$

then: [1] $E(X) = \mu$

[2] $\text{Var}(X) = \sigma^2$

$$P(X > \mu) = 1/2$$

$$P(X \leq \mu) = 1/2$$

$$P(X = \mu) = \text{zero}$$

31-10-2017

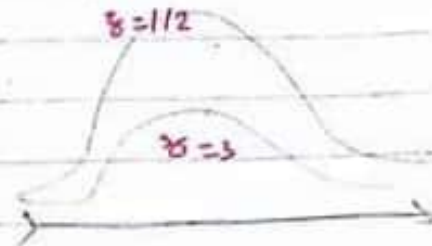
⑥

* Note

$$X: N(\mu, \sigma^2)$$

→ σ is large $\Rightarrow$ the curve is short and wide

→ σ is small $\Rightarrow$ The curve is tall and narrow



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Defn

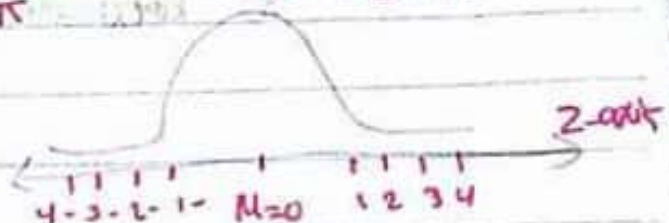
$$X = N(\mu, \sigma^2)$$

In the normal distribution when $\mu=0$ and $\sigma^2=1$

the distribution is called "Standard Normal distribution"

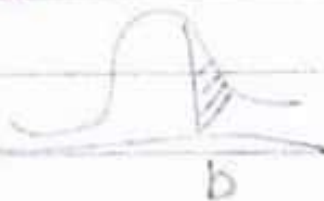
$$Z = N(0, 1)$$

$$\text{p.d.f} \Rightarrow f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \quad -\infty < z < \infty$$



Fact If $Z: N(0, 1)$ then:

1. $P(Z < a)$ = the area to the left of a
2. $P(Z > b)$ = The area to the right of b
3. $P(Z = 0) = 0$



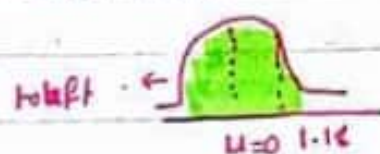
⑦

$$\# P(a \leq X \leq b) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \cdot dz$$

$$\# P(Z < a) = \int_{-\infty}^{\infty} P(z) \cdot dz$$

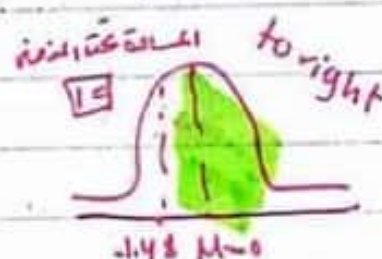
Ex: In the standard normal Curve find:

1. $P(Z \leq 1.18) = 0.8810$



z	0.08
1.1	0.8810

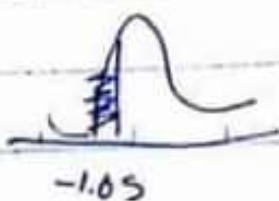
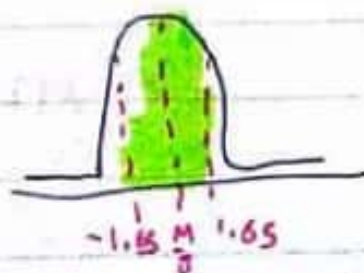
[2] $P(Z > -1.48)$
 $= 1 - P(Z < -1.48)$
 $= 1 - 0.0694 = 0.9306$



In general

$$P(Z > a) = 1 - P(Z \leq a)$$

[3] $P(-1.65 < Z < 1.65)$
 $= 0.9505 - 0.0495$
 $= 0.9010$



$$P(a < Z < b) = P(Z < b) - P(Z < a)$$

②

Ex find z_0 such that:

1. $P(Z > z_0) = 0.1567$

$\downarrow$
 $z_0 = 1$

z	0.00
-1.0	0.1567

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مقدار z

z 0.500 < z < 0.9998

الموجب 0.00 < z < 0.499

مثال - 12

2. $P(Z < z_0) = 0.900$

مستوي ثقة من 0.900 بينه وبينه القيمة الحرجة والتوزيع

0.8997 ← 0.900 → 0.9015
.003 .015

أقرب

$z_0 = 1.28$

3) $P(Z > z_0) = 0.2$

$1 - P(Z < z_0) = 0.2$

$P(Z < z_0) = 0.800$

0.7995 0.800 0.8023
.0005 .0023

أقرب

$z_0 = 0.84$

Result :-

If $X \sim N(\mu, \sigma^2)$ then $Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$
Normal distribution Standard normal

Ex. let X has a normal distribution with Mean of 60
Variance of 100 $X \sim N(\mu=60, \text{Var}^2=100)$

Find:

[1] $P(X > 80)$?

$$P(X > 80) = P\left(\frac{X - \mu}{\sigma} > \frac{80 - 60}{10}\right) = P(Z > 2)$$

$$= 1 - P(Z < 2)$$

$$= 1 - 0.9772$$

$$= 0.0228$$

[2] $P(60 < X < 80) = P\left(\frac{60 - 60}{10} < \frac{X - \mu}{\sigma} < \frac{80 - 60}{10}\right)$

$$= P(0 < Z < 2)$$

$$= P(0 < Z < 2)$$

$$= P(Z < 2) - P(Z < 0)$$

$$= 0.9772 - 0.5$$

$$= 0.4772$$

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2-11-2017

(10)

13-10 $X \sim N(M, \sigma^2)$

$$P.d.f = P(X) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(X-M)^2}$$

17-10 $P.d.f = f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$

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Application :

Ex "1" Score on a class exam have a Mean of 85 and standard deviation of 5

let X represents student test result on the exam

Assume X is a random normal variable.

$$X \sim N(M=85, \sigma^2=25)$$

What is the prob a randomly selected student will get a grade of

(a) lower than 90

Soln $P(X < 90) \rightarrow P\left(\frac{X-M}{\sigma} < \frac{90-85}{5}\right) = P(Z < 1)$



$$= 0.8413$$

(b) higher than 85

$$P(X > 85) = P\left(\frac{X-M}{\sigma}, \frac{85-85}{5}\right) = P(Z > 0)$$

$$\begin{aligned} &= 1 - P(Z < 0) \\ &= 1 - 0.5 \\ &= 0.5 \end{aligned}$$

② between 90, 95?

$$P(90 < x < 95) = P\left(\frac{90-85}{5} < \frac{x-\mu}{\sigma} < \frac{95-85}{5}\right)$$

$$= P(1 < Z < 2)$$

$$= P(Z < 2) - P(Z < 1)$$

$$= 0.9772 - 0.8413$$

$$= 0.135$$

④ Higher than 95 or less than 75?

Soln $P(X > 95 \text{ or } X < 75)$

$$= P(X > 95) + P(X < 75)$$

$$= P\left(Z > \frac{90-85}{5}\right) + P\left(Z < \frac{75-85}{5}\right)$$

$$= P(Z > 2) + P(Z < -2)$$

$$= 1 - P(Z > 2) + P(Z < -2) \quad \text{not}$$

$$1 - 0.9772 + 0.0228 = 0.0456$$

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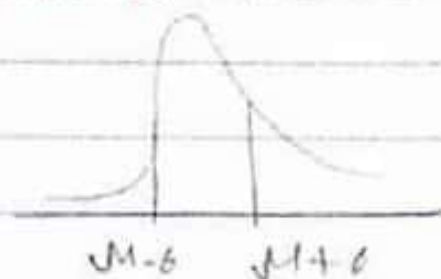
9.51

⑤ within one standard deviation of the Mean?

$$P(\mu - \sigma < X < \mu + \sigma)$$

$$P(85-5 < X < 85+5)$$

$$P(80 < X < 90)$$



Find the 90th Percentile?

$$P_{90} : P(X < P_{90}) = 0.9$$

$$C = ??$$

$$\text{let } c = P_{90}$$

$$\Rightarrow \frac{90}{100} = 0.9$$

(10)

$$P(X \leq C) = 0.9$$

$$P\left(\frac{X - \mu}{\sigma} \leq \frac{C - 85}{5}\right) = 0.9$$

$$P(Z \leq Z_0) = 0.9$$

$$\text{Where } Z_0 = \frac{C - 85}{5}$$

$$Z_0 = 1.28$$

$$1.28 = \frac{C - 85}{5}$$

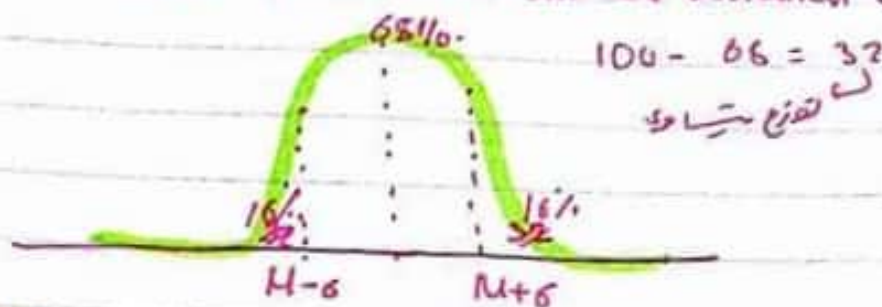
$$C = 85 + 5(1.28) = 91.4$$

H.W P_{78} ? $P(X < P_{78}) = 0.78$

(13)

Ex 2" let $X \sim N(\mu, \sigma^2)$

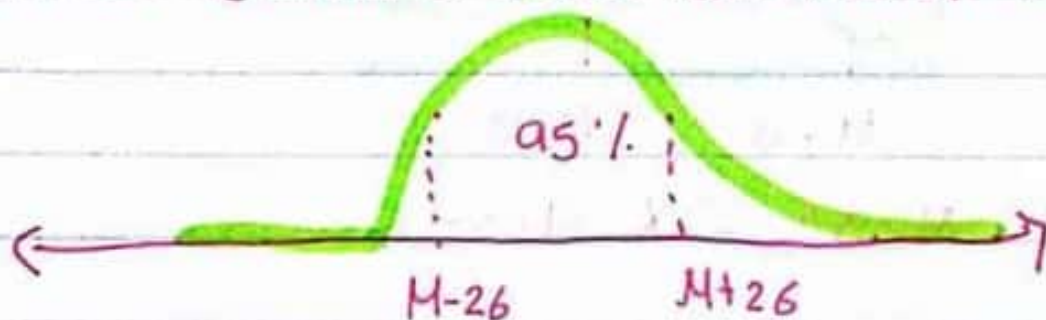
1. About 68% of the observation are between $\mu - \sigma$ & $\mu + \sigma$ (within the standard deviation of the Mean)



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$$\begin{aligned}
 \# \quad & P(\mu - \sigma < X < \mu + \sigma) = ? \\
 & = P\left(\frac{\mu - \sigma - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{\mu + \sigma - \mu}{\sigma}\right) \\
 & = P(-1 < Z < 1) \\
 & = P(Z < 1) - P(Z < -1) \\
 & = 0.6826.
 \end{aligned}$$

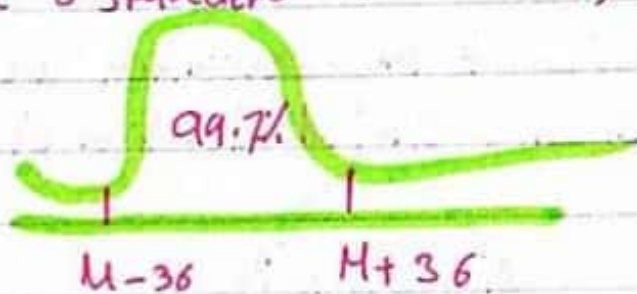
- [2] about of 95% of the observation are between $\mu - 2\sigma$ & $\mu + 2\sigma$ (within 2 standard deviation of the Mean).



$$P(\mu - 2\sigma < X < \mu + 2\sigma).$$

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[3] about 99.7% of the observation are between $\mu - 3\sigma$ and $\mu + 3\sigma$ (within 3 standard).



$$P(\mu - 3\sigma < X < \mu + 3\sigma)$$

H.w

(15)

① if the X is normal with Mean = 2
 $P(X > 10) = 0.1587$
Find the standard deviation?

② let $X \sim N(\mu, \sigma^2)$

If $P(X < 45) = 0.1587$, $P(X > 60) = 0.0228$
Determine μ , σ^2 ?

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(16)

* Sampling Distribution

* We have finite (infinite) population with unknown μ & σ^2

Main study is to be estimate μ (Population Mean)

We select random sample of size n from population

$X_1, X_2, \dots, X_n$

$$\text{Sample Mean} = \bar{X} = \frac{X_1 + X_2 + X_3 + \dots + X_n}{n} \quad (X_i \text{ is indep})$$

Population Mean $\mu \rightarrow$ unknown

[$\bar{X}$ is an estimator μ] $\bar{X} \sim \mu$

Q What is the distribution of $\bar{X}$;
Assume $X_1, X_2, X_n, \dots$ from $N(\mu, \sigma^2)$

Soln $\square E(\bar{X}) = E\left(\frac{X_1 + X_2 + X_n}{n}\right) = \frac{1}{n} \times E(X_1) + E(X_2) + \dots$

$$= \frac{1}{n} (\mu + \mu + \mu)$$
$$= \frac{1}{n} \cdot n(\mu)$$
$$= \mu$$

$E(\bar{X}) = \mu$

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(17)

$$\begin{aligned}
 \text{[2]} \quad \text{Var}(\bar{X}) &= \text{Var}\left(\frac{X_1 + \dots + X_n}{n}\right) \\
 &= \frac{1}{n^2} (\text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)) \\
 &= \frac{1}{n^2} (\sigma^2 + \sigma^2 + \dots + \sigma^2) \\
 &= \frac{1}{n^2} \cdot n \sigma^2 = \boxed{\frac{\sigma^2}{n}}
 \end{aligned}$$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

$\text{Var}(X_1 + X_2) \neq \text{Var}(X_1) + \text{Var}(X_2)$
 unless X_1 & X_2 are **Indep**.

Result:

$$\begin{aligned}
 &\bar{X} : N(M, \sigma^2) \\
 * \quad X &\rightarrow Z = \frac{X - M}{\sigma} \rightarrow Z \text{ table} \quad Z : N(0, 1)
 \end{aligned}$$

$$\bar{X} \rightarrow Z = \frac{\bar{X} - M}{\sigma} \rightarrow Z \text{ table} \quad Z \sim N(0, 1)$$

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(18)

Ex : Suppose a random sample of size $n=25$ is selected from a population that is normally dist with Mean = 106 & $\sigma=12$ find?

a The Mean & S.P of the sampling dist of sample Mean ($\bar{X}$) ?

b find $P(\bar{X} - M < 4)$?

Soln **a** $E(\bar{X}) = M = 106$

$$\text{S.P of } (\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{12}{\sqrt{25}} = \frac{12}{5} = 2.4$$

$$\begin{aligned} \text{b} \quad P(\bar{X} - M < 4) &= P\left(\frac{\bar{X} - M}{\sigma/\sqrt{n}} < \frac{4}{\sigma/\sqrt{n}}\right) \\ &= P\left(Z < \frac{4}{12/5}\right) \end{aligned}$$

$$= P(Z < 1.64)$$

$$= 0.9525$$

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Central Limit Theorem (CLT)

let $x_1, x_2, \dots, x_n$ be random sample from any distribution with Mean μ and σ^2 then: For large n (1a)
 $n > 30$

1) $E(\bar{x}) = \mu$

2) $\text{Var}(\bar{x}) = \sigma^2 / n \rightarrow [n \text{ increase} \Rightarrow \text{Var}(\bar{x}) \text{ decrease}]$

3) $\bar{x} \underset{\text{approximately}}{\sim} N(\mu, \frac{\sigma^2}{n})$

approximately.

$\rightarrow Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \underset{\text{بالقريب}}{\sim} N(0,1)$

Ex

Note $\sum_{i=1}^n x_i = x_1 + x_2 + \dots + x_n = S_n$ (Sample Sum)

$S_n \sim N(n\mu, n\sigma^2)$

or $P\left(\sum_{i=1}^n x_i > \frac{5}{4}\right) = P(\bar{x} > 5/4)$
 prob (4)

Ex Suppose that random sample of size $n=100$ is drawn from population with $\mu=70$

$\sigma = 20$

what is the prob that the sample mean will

① ~~Mean~~ More than 70 ? $P(\bar{x} > 70)$

② less than 73?

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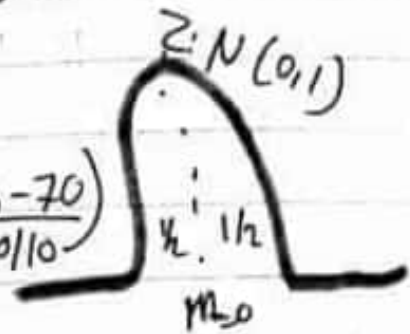
(20)

By CLT :-

$$\textcircled{a} P(\bar{X} > 70) \approx P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{70 - \mu}{\sigma/\sqrt{n}}\right) \\ = P(Z > 0) = 0.5$$

$$\textcircled{b} P(\bar{X} < 73) \approx P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{73 - 70}{20/10}\right)$$

$$P(Z < 1.51) = 0.9332$$



$$P(Z < 0) = 0.5$$

$$P(Z > 0) = 0.5$$

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ExSuppose $\mu = 10$ of orange boxes
 $\sigma = 2$ $n = 100$ (A car with threshold = 1000 Kg)

What is the prob that car will break down?

Soln let $x_1, x_2, \dots, x_n$ (weight of box)

$$P\left(\sum_{i=1}^{100} x_i > \frac{1000}{100}\right) = P(\bar{X} > 10) \approx$$

$$P\left(Z > \frac{10 - 10}{2/\sqrt{100}}\right) = P(Z > 0) = 0.5$$

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or $S_n \approx N(nM, \sigma^2)$ (21)

$$P(S_n > 1000)$$

$$P\left(\frac{S_n - nM}{\sqrt{n\sigma^2}} > \frac{1000 - nM}{\sqrt{n\sigma^2}}\right)$$

* 8-11-2017

* Result :-

① IF $X \sim N(M, \sigma^2)$ then $Z = \frac{X - M}{\sigma} \sim N(0,1)$

② (CLT)
infinite population $\begin{cases} E(\bar{x}) = M_{\bar{x}} = M \\ \text{Var}(\bar{x}) = \sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} \end{cases}$

$$\bar{x} \approx N\left(M, \frac{\sigma^2}{n}\right) \Rightarrow Z = \frac{\bar{x} - M}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$$

$$S_n = \sum_{i=1}^n x_i \sim N(nM, n\sigma^2) \Rightarrow Z = \frac{S_n - nM}{\sqrt{n}\sigma} \sim N(0,1)$$

Ex IF $X \sim N(60, 100)$ if the sample size $n=100$
Find each of the following :-

- [a] $P(X < 68)$?
- [b] $P(\bar{x} > 59)$
- [c] $P(58 < \bar{x} < 60)$
- [d] $P(\bar{x} = 60)$
- [e] $P\left(\sum_{i=1}^{100} x_i < 6100\right)$?

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(22)

Soln

$$\begin{aligned} \text{a)} &= P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} < \frac{68 - 60}{10}\right) = P(Z < 0.8) \\ &= 0.7881 \end{aligned}$$

$$\begin{aligned} \text{b)} P(\bar{X} > 59) &= P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} > \frac{59 - 60}{10/\sqrt{100}}\right) \\ &= P(Z > -1) = 1 - P(Z < -1) = 0.8413 \end{aligned}$$

H.w

$$\begin{aligned} \text{d)} P(\bar{X} = 60) &= 0 \\ 60 \leq \bar{X} &\leq 60 \end{aligned}$$

$$\text{e)} P\left(\sum_{i=1}^{100} x_i < 6100\right) = P\left(\frac{\sum_{i=1}^{100} x_i - n\mu}{\sqrt{n}\sigma} < \frac{6100 - 100(60)}{10 \cdot 10}\right)$$

$$\text{H.w} = P \dots$$

$$\text{or} = P\left(\sum_{i=1}^{100} \frac{x_i}{100} < \frac{6100}{100}\right) = P(\bar{X} < 61)$$

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H.W

- ① the length of the trees in a farm is normally dist with
Mean = 42 & $\sigma = 4$, $n = 9$

(sample Mean)

- (a) what the prob that have an average of length less than 43?
(b) What is the prob that 4-trees have total length greater than 180? $P(x_1 + x_2 + x_3 + x_4 > 180)$

[2] a random sample of size 25 is selected at random from a normal dist with mean of μ variance of 16? find

① $P(16 < \bar{x} < 20)$

② $P(\sum_{i=1}^{25} x_i < 525)$

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(25)

Bernolli distribution :-

Def: Bernolli trial is any trial or experiment with only two possible outcomes $\rightarrow$ success
 $\rightarrow$ failure

$$X = \begin{cases} 1 & : \text{Success with prob } (P) \\ 0 & : \text{Failure with prob } (1-P) \end{cases}$$

P.M.F :

X	0	1	Sum
$P(X)$	$1-P$	P	1
$X \cdot P(X)$	0	P	P
$X^2 \cdot P(X)$	0	P	P

$$\Rightarrow \sum x \cdot P(X) = E(X)$$

E(X)

$$\Rightarrow \sum x^2 \cdot P(X) = E(X^2)$$

$$\# E(X) = \mu = \sum x \cdot P(X) \Rightarrow \mu = P$$

$$\# \sigma^2 = E(X^2) - (E(X))^2$$

$$= P - P^2 = P(1-P) \Rightarrow \sigma^2 = P(1-P)$$

X: Bernolli (P).

X: B(n, P)

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Relation between Bernolli and Binomial

A sequence of n-Bernolli trials (Independent)
 . (P fixed)

$$S_n = X_1 + X_2 + X_3 + \dots + X_n = \text{Sum of } n \text{ Indep Bernolli} \\ = X: B(n, P)$$

by (CLT), for large n or P close to 1/2

S_n approx by $N(n\mu, n\sigma^2)$

$S_n \approx N(n\mu, n\sigma^2)$

$$X: N(n, P) \approx N(n\mu, n\sigma^2)$$

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Binomial $E(x) = np = nH \Rightarrow \boxed{H=P}$

$Var(x) = np(1-p) = n\sigma^2 \Rightarrow \sigma^2 \neq P(1-p)$

Tip

$X: B(np, P) \approx N(np, np(1-p))$

Binomial approximated by normal

Discrete

Continuous

$a \leq x \leq b$

$a-0.5 \leq x \leq b+0.5$

$x=a$

$a-0.5 < x < b+0.5$

$x \leq b$

$x \leq b+0.5$

$x \geq a$

$x \geq a-0.5$

$X: B(np, P)$

$P(a \leq x \leq b) = P(a-0.5 \leq x \leq b+0.5)$

$= P\left(\frac{a-0.5-np}{\sqrt{np(1-p)}} \leq \frac{x-np}{\sqrt{np(1-p)}} \leq \frac{b+0.5-np}{\sqrt{np(1-p)}}\right)$

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9/11/2017

(27)

Ex Find $\int_{-\infty}^{\infty} e^{-z^2/2} dz$ $Z = (N(0,1))$

Recall: (Binomial Approx imdted by normal For large n/p
p's Close to 1/2.

$$X: B(n, p) \approx N(np, np(1-p))$$

$$Z = \frac{X - np}{\sqrt{np(1-p)}} : N(0,1)$$

27

Page 43
Ex 27

Ex Suppose we toss a fair coin 20 times

let $x = \#$ of heads

$$X: B(n=20, p=1/2)$$

Q: Find $P(9 < x < 11)$? $P(x=9) + P(x=10) + P(x=11)$

Soln

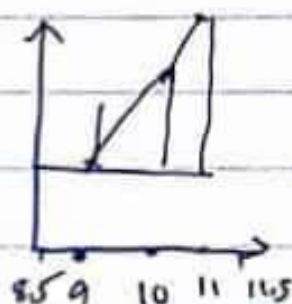
Exact Soln = $\sum_{k=9}^{11} \binom{20}{k} \left(\frac{1}{2}\right)^k \left(1 - \frac{1}{2}\right)^{20-k} = \boxed{0.25}$

$$P(9 < x < 11) \approx P(8.5 \leq x \leq 11.5)$$

↳ Continuity correction:

$$= P\left(\frac{8.5 - np}{\sqrt{np(1-p)}} \leq \frac{x - np}{\sqrt{np(1-p)}} \leq \frac{11.5 - np}{\sqrt{np(1-p)}}\right)$$

$$= np = 20 \times \frac{1}{2} = 10$$



$$np(1-p) = 20 \times \frac{1}{2} \times \frac{1}{2} = 5$$

(28)

Ex let $X = B(15, 0.6)$: $n=15$
 $p=0.6$

Use normal approx to a binomial dist to find
 $P(X=8)$?

Exact Soln : $P(X=8) = \binom{15}{8} (0.6)^8 (0.4)^7 = 0.1771$

Calculation of $P(X=8) = P(7.5 \leq X \leq 8.5) \approx P\left(\frac{7.5-9}{\sqrt{3.6}} \leq Z \leq \frac{8.5-9}{\sqrt{3.6}}\right)$
 $nP=9$
 $nP(1-p)=3.6$

$\leq \frac{8.5-9}{\sqrt{3.6}}$

$= P(-1.39 < Z < -0.28) = 0.1413$

H.W Use normal Approx to a Binomial dist with $n=10$
to find $P(3 \leq X \leq 7)$? $p=0.5$

28

Chapter '5'

(29)

1. We have infinite Normal population whose Mean is μ and Variance is σ^2 $X \sim N(\mu, \sigma^2)$



$\downarrow$ unknown

Random Sample of size n

$\bar{X}$ = sample Mean is a good estimate of population Mean

$$\bar{X} \rightarrow E(\bar{X}) = \mu_{\bar{X}} = \mu$$

$$\bar{X} \rightarrow \text{Var}(\bar{X}) = \sigma_{\bar{X}}^2 = \sigma^2/n$$

2. We have finite population of size N whose Mean is μ and Variance σ^2

$\hookrightarrow$ Random Sample of size n (without replacement)

$$\bar{X} \rightarrow E(\bar{X}) = \mu_{\bar{X}} = \mu$$

$$\bar{X} \rightarrow \text{Var}(\bar{X}) = \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right) \rightarrow \text{Correction factor}$$

Remark.

1. The standard deviation of $\bar{X}$ ($S.D(\bar{X})$) is called "standard error" of the mean ($S.E(\bar{X})$)

$$S.E(\bar{X}) = \begin{cases} \sigma_{\bar{X}} = \sigma/\sqrt{n} & \text{for infinite population} \\ \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} & \text{for finite population size.} \end{cases}$$

standard error

2. The Standard Error $S.E(\bar{X})$ $\uparrow$ as $\sigma \uparrow$
 $\downarrow$ As $n \uparrow$
 $\hookrightarrow$ sample space.

(30)

Ex A Statistic class has six student ages areAges 18, 18, 19, 20, 20, 21 $\Rightarrow N = 6$ Sample of size 2 ($n = 2$)

1. Find the number of samples

$$\binom{N}{n} = \binom{6}{2} = \frac{6!}{2!4!} = 15$$

2. Find the Population Mean & Population Variance?

30

Soln $M = \frac{18+18+19+20+20+21}{6} = \frac{58}{3} \approx 19.33$

$$\sigma^2 = \frac{\sum_{i=1}^6 (X_i - M)^2}{N} \approx 1.22$$

X_i	$(X_i - \bar{x})$	$(X_i - \bar{x})^2$

3. Find the Mean (Expected value) of Variance (C) Standard Deviation of the sample Mean ($\bar{x}$) = ?

Soln $E(\bar{x}) = M_{\bar{x}} = M = 19.33$

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right) = \frac{1.22}{2} \left(\frac{6-2}{6-1} \right) = 0.488$$

$$(C) = S.D(\bar{x}) = S.E(\bar{x}) = \sqrt{0.488}$$

H.W A random Sample of size $n = 64$ is selected at random from finite population of size 250

the mean & Variance of this population are 16, 4

Find the Standard error of the sample?

12 / 11 / 2017

chapter "6"

Estimation

(31)

Defn:

1. A **Statistic**: is a statistical measure computed for **Sample**.
2. A **parameter**: is a statistical measure computed for population (is a property of **population**).

Statistic	Parameter
$\bar{x}$	μ
s^2	σ^2

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An estimator is statistic used to estimate unknown
 $\downarrow$
 Parameter.

$\bar{x}$ is an estimator for μ
 $\downarrow$
 $\downarrow$
 Statistic Parameter

Result:

Let $X_1, X_2, \dots, X_n$ be a random sample from normal population with mean μ & variance σ^2 $X: N(\mu, \sigma^2)$

1. $\bar{X}$ is unbiased estimator for μ $E(\bar{X}) = \mu$

2. $\bar{X}$ is a consistent estimator for μ

$$\lim_{n \rightarrow \infty} \bar{X} = \mu$$

① $\hat{\theta}$ unbiased for θ
 estimator.

$$E(\hat{\theta}) = \theta$$

② $\hat{\theta}$ consistent for θ

$$\lim_{n \rightarrow \infty} \hat{\theta} = \theta$$

(32)

3. $\bar{X} : N(M, \sigma^2/n)$

4. S^2 is unbiased estimator for σ^2 , $(E(S^2) = \sigma^2)$

* Remark: Estimation is Either

① Point estimate [$\bar{X}$ is an estimator for M]

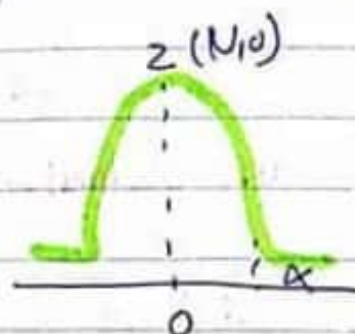
② $(1-\alpha) 100\%$ Confidence Interval (C.I)

$(1-\alpha) 100\%$ C.I for θ is Interval (L, U)

Computed using random sample such that $P(L < \theta < U) = 1-\alpha$

* Notation

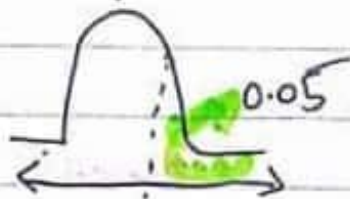
Z_α : the Value where $P(Z > Z_\alpha) = \alpha$



Determine:

1. $Z_{0.05}$?

$1 - 0.05 = 0.9500$ $z = 0.05$



32

0.9495 ← 0.9500 → 0.9505

5 5

1.64 $Z_{0.05} = 1.64$

في طاقه
مناقصه التوزيع

2. $Z_{0.025}$? $1 - 0.025 = 0.9750$

$z_{0.025} = 1.96$

3. $Z_{0.005}$? $1 - 0.005 = 0.995$

0.9949 ← 0.9950 → 0.9951

1 1

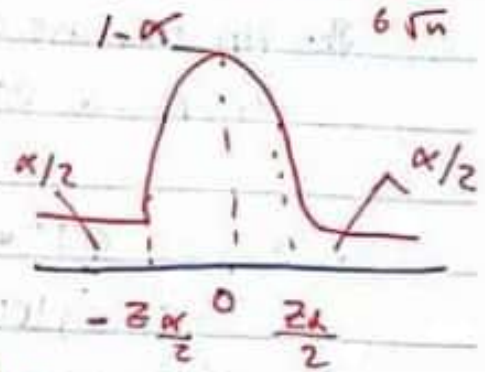
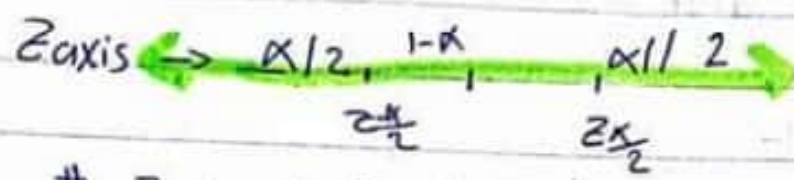
$2.57 = Z_{0.005}$

(33)

C.I for Mean (μ) unknown.

Let $x_1, x_2, \dots, x_n$ be a random sample from $X: N(\mu, \sigma^2)$ normal population.
Given the confidence coefficient $1 - \alpha$

$$P\left(-\frac{Z_{\alpha}}{2} < Z < \frac{Z_{\alpha}}{2}\right) = 1 - \alpha$$



$$\# -Z_{\alpha/2} < Z < Z_{\alpha/2}$$

$$-Z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < Z_{\alpha/2}$$

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$$-Z_{\alpha/2} \cdot \sigma/\sqrt{n} < \bar{X} - \mu < Z_{\alpha/2} \cdot \sigma/\sqrt{n}$$

$$|\bar{X} - \mu| < [Z_{\alpha/2} \cdot \sigma/\sqrt{n}]$$

E maximum error is estimating μ by $\bar{X}$

$$E = Z_{\alpha/2} \cdot \sigma/\sqrt{n} = Z_{\alpha/2} \cdot S.E(\bar{X})$$

Back to * $\rightarrow \bar{X} - Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$

$$\bar{X} - E < \mu < \bar{X} + E$$

$$\bar{X} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

Point

Estimate $\bar{X} \pm Z_{\alpha/2} \cdot \sigma/\sqrt{n}$ standard error

Confidence Multiplier

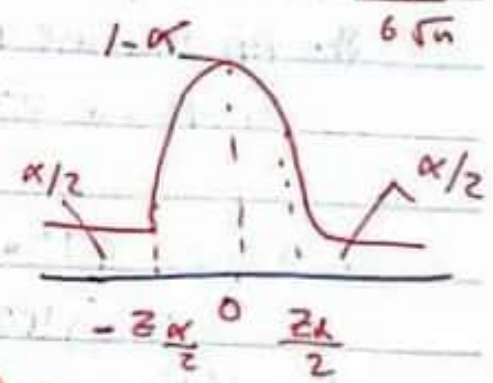
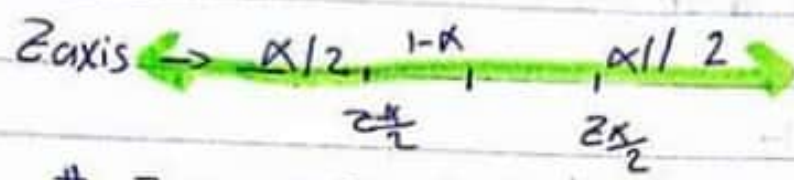
margin of error.

(33)

C.I for Mean (μ) unknown.

Let $x_1, x_2, \dots, x_n$ be a random sample from $X: N(\mu, \sigma^2)$ normal population.
Given the confidence coefficient $1-\alpha$

$$P\left(-\frac{Z_{\alpha}}{2} < Z < \frac{Z_{\alpha}}{2}\right) = 1-\alpha$$



$$\# Z_{-\alpha/2} < Z < Z_{\alpha/2}$$

$$-Z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < Z_{\alpha/2}$$

33

$$-Z_{\alpha/2} \cdot \sigma/\sqrt{n} < \bar{X} - \mu < Z_{\alpha/2} \cdot \sigma/\sqrt{n}$$

$$|\bar{X} - \mu| < [Z_{\alpha/2} \cdot \sigma/\sqrt{n}]$$

E maximum error is estimating μ by $\bar{X}$

$$E = Z_{\alpha/2} \cdot \sigma/\sqrt{n} = Z_{\alpha/2} \cdot S.E(\bar{X})$$

Back to * $\rightarrow \bar{X} - Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$

$$\bar{X} - E < \mu < \bar{X} + E$$

$$\bar{X} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

Point

Estimate $\bar{X} \pm Z_{\alpha/2} \cdot \sigma/\sqrt{n}$ standard error

Confidence Multiplier

margion of error.

(30)

Note

$$1. E = \frac{Z_{\alpha}}{2} \cdot \frac{\sigma}{\sqrt{n}}$$

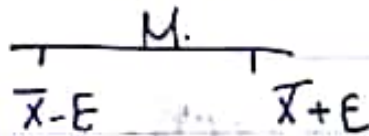
* An increase $n \rightarrow$ (sample size) error estimation decrease
is $n \rightarrow \infty \Rightarrow \text{error} = 0$

$$2. \bar{X} - E < M < \bar{X} + E$$

↓ $(1-\alpha)100\%$ C.I for M

A. $P(M(\text{unknown}) \in [\bar{X} - E, \bar{X} + E]) = (1-\alpha)100\%$

B. Any point in Interval $I = [\bar{X} - E, \bar{X} + E]$
is considered As an estimate of M



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* 15-11-2017

Confidence Interval For Population Mean :

We have a population with unknown Mean M .

$\bar{X}$ → point estimate for M .

Interval estimate

Case I : σ : Known

* $(1-\alpha)\%$ CI estimate for M .

$$\bar{X} - \left(\frac{Z_{\alpha}}{2} \cdot \frac{\sigma}{\sqrt{n}} \right) < M < \bar{X} + \left(\frac{Z_{\alpha}}{2} \cdot \frac{\sigma}{\sqrt{n}} \right)$$

lower limit upper limit

(36)
 # A the maximum error of this estimate for μ is :-

$$E = \frac{Z_{\alpha/2} \cdot \sigma}{\sqrt{n}} = \frac{Z_{\alpha/2} \cdot S.E(\bar{x})}{1}$$

* Sample size of estimating μ :

$$E = \frac{Z_{\alpha/2} \cdot \sigma}{\sqrt{n}}$$

$$E \cdot \sqrt{n} = \sigma \cdot \frac{Z_{\alpha/2}}{1}$$

$$n = \left(\frac{\sigma \cdot Z_{\alpha/2}}{E} \right)^2$$

$$\sqrt{n} = \frac{\sigma \cdot Z_{\alpha/2}}{E}$$

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Ex(1) : A random size $n=25$ is taken from normal population with unknown μ and $\sigma^2 = 225 \Rightarrow \sigma = 15$

If the sample Mean is found to be 13.8 construct 90% C.I for μ

Soln $1 - \alpha = 0.9 \Rightarrow \alpha = 0.1 \Rightarrow \frac{Z_{\alpha}}{2} = \frac{Z_{0.1}}{2} = Z_{0.05} = 1.64$

$$\bar{x} - E < \mu < \bar{x} + E \quad \bar{x} = 13.8$$

$$E = \frac{Z_{\alpha/2} \cdot \sigma}{\sqrt{n}} = \frac{1.64 \cdot 15}{5} = 1.64 \times 3 = 4.92$$

$$13.8 - 4.92 < \mu < 13.8 + 4.92$$

$$\mu \in [8.88, 18.72]$$

(36)

Ex:2 To be confidence 95% that the sampling error in estimating μ by $\bar{x}$ is 5.

Assuming $\sigma = 25$ what is the required sample size?

$$1 - \alpha = 0.95$$

$$\alpha = 0.05$$

Soln $n = \left(\frac{Z_{\alpha/2} \cdot \sigma}{E} \right)^2$

$$Z_{\alpha/2} = Z_{0.025} = 1.96$$

$$n = \left(\frac{1.96 \cdot 25}{5} \right)^2$$

$$n = (9.8)^2 = 96.04 \text{ round to the next higher integer}$$

$$n = 97$$

Case II: σ unknown, n : large ($n \geq 30$)

S replacement $\rightarrow \sigma$
of

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

* $(1 - \alpha) 100\%$ C.I estimate for μ :

$$\bar{x} - \left(Z_{\alpha/2} \cdot \frac{S}{\sqrt{n}} \right) < \mu < \bar{x} + \left(Z_{\alpha/2} \cdot \frac{S}{\sqrt{n}} \right)$$

** the maximum error of this estimate $\mu \Rightarrow E = Z_{\alpha/2} \cdot \frac{S}{\sqrt{n}}$

Sample size for estimating for $\mu \Rightarrow n = \left(\frac{Z_{\alpha/2} \cdot S}{E} \right)^2$

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(37)

Case III : σ : unknown, n : small ($n < 30$)

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$\rightarrow$ t-distribution with $(n-1)$ degree of freedom (v)

$t_{\alpha, v}$?

~~$t_{\alpha, v}$~~ ?

$df = v$

Ex: find $t_{0.025, 7}$?

$$\alpha = 0.025$$

$$t_{0.025, 7} = 2.365$$

$$v = 7$$

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* $(1-\alpha)100\%$ C-I Estimate for $\mu =$

$$\bar{x} - \left(t_{\frac{\alpha}{2}, n-1} \cdot \frac{s}{\sqrt{n}} \right) < \mu < \bar{x} + \left(t_{\frac{\alpha}{2}, n-1} \cdot \frac{s}{\sqrt{n}} \right)$$

the Maximum error of this Estimate

$$E = t_{\frac{\alpha}{2}, n-1} \cdot \frac{s}{\sqrt{n}}$$

Ex: Consider the following Sample 3, 6, 12, 4, 6, 8, 10 taken randomly from unknown population

Find:

① The (point estimate) of the population Mean (μ)?

Soln $\bar{x} = \frac{\sum_{i=1}^7 x_i}{n} = \frac{79}{7} = 11.29$

② the point estimate of the population variance (σ^2)?

Soln $s^2 = \frac{\sum_{i=1}^7 (x_i - \bar{x})^2}{7-1} = \frac{62}{6} = 10.33$

(38)

3. the point estimate of σ ? $S = \sqrt{10.33}$ 4. Find the ~~estim~~ standard error of the Sample Mean

Soln $S.E(\bar{x}) = \frac{S}{\sqrt{n}} = \frac{\sqrt{10.33}}{\sqrt{7}}$ $E_{\frac{\alpha}{2}, n-1} \cdot S.E(\bar{x})$ σ (unknown)
 $n=7 < 30 \rightarrow \sigma$ replasment

$$S.E(x) = 1.2$$

5. Find 90% C.I for μ

$$* t_{\frac{\alpha}{2}, n-1} = t_{0.05, 6} = 1.943$$

$$1 - \alpha = 0.9 \Rightarrow \boxed{\alpha = 0.1}$$

$$\bar{x} \pm \left(t_{\frac{\alpha}{2}, n-1} \cdot \frac{S}{\sqrt{n}} \right)$$

$$7 \pm (1.943 * 1.2)$$

$$7 - 2.331 < \mu < 7 + 2.331$$

$$\mu \in [4.61, 9.39]$$

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Recall that:

Population with unknown Mean (μ)

point estimate

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Interval

$(1-\alpha) 100\%$ C.I For μ .

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σ : Known

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$\bar{x} \pm z_{\frac{\alpha}{2}} \left(\frac{\sigma}{\sqrt{n}} \right)$$

E

$n \gg 30$ large

replace σ by s

$$z = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$\bar{x} \pm z_{\frac{\alpha}{2}} \left(\frac{s}{\sqrt{n}} \right)$$

E

σ unknown

$n < 30$ small

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$\bar{x} \pm t_{\frac{\alpha}{2}, n-1} \left(\frac{s}{\sqrt{n}} \right)$$

E

$$* n = \left(\frac{z_{\frac{\alpha}{2}} \cdot \sigma}{E} \right)^2$$

$1-\alpha = 0.9$
 $\alpha = 0.1$

90% C.I $\Rightarrow z_{\frac{\alpha}{2}} = z_{0.05} = 1.64$ find

95% C.I $\Rightarrow$

99% C.I $\Rightarrow$

find

Estimation of the population (P)

Population has a proportion (P). $0 < P < 1$

We want study about (P)

We consider a sample of size n from Bernoulli (P)

$X_i = \begin{cases} 1 & \text{with Prob} = p \text{ (Success)} \end{cases}$

$\begin{cases} 0 & \text{with Prob} = 1-p \text{ (Failure)} \end{cases}$

(40)

$$X_1 + X_2 + \dots + X_n = X(n, p) \begin{cases} E(X) = np \\ \text{Var}(X) = np(1-p) \end{cases}$$

= total number of successes.

* Population Proportion $P = \frac{\# \text{ of success in the population}}{N}$

* Sample Proportion $\hat{p} = \frac{\# \text{ of successes in the sample}}{n}$

$\hat{p}$ Point estimator $\rightarrow P$

$$(\hat{p} = \frac{X}{n}) \quad \text{عدد النجاحات في العينة}$$

Result:

$$1. E(\hat{p}) = E\left(\frac{X}{n}\right) = \frac{1}{n} E(X) = \frac{1}{n} np = p$$

$E(\hat{p}) = P$ $\hat{p}$ is unbiased estimator for P

$$2. \text{Var}(\hat{p}) = \text{Var}\left(\frac{X}{n}\right) = \frac{1}{n^2} \text{Var}(X) = \frac{1}{n^2} np(1-p) = \frac{P(1-P)}{n}$$

$$3. S.E(\hat{p}) = \sqrt{\frac{P(1-P)}{n}}$$

For Large (n) Using CLT:

$$\hat{p} \approx N\left(P, \frac{P(1-P)}{n}\right)$$

$\mu = E(\hat{p}) \quad \sigma^2 = \text{Var}(\hat{p})$

$$Z = \frac{\hat{p} - P}{\sqrt{\frac{P(1-P)}{n}}} \sim N(0, 1)$$

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$$

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$$\left\{ \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \right\} - z_{\frac{\alpha}{2}} < z < z_{\frac{\alpha}{2}}$$

$$p - z_{\frac{\alpha}{2}} \cdot \sqrt{\frac{p(1-p)}{n}} < \hat{p} < p + z_{\frac{\alpha}{2}} \cdot \sqrt{\frac{p(1-p)}{n}}$$

replace p by $\hat{p}$

$$\hat{p} - \left(z_{\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right) < p < \hat{p} + \left(z_{\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right)$$

E

Population proportion (p)
Interval

Point estimate of p

$(1-\alpha)100\%$ C.I

$$\hat{p} = \frac{x}{n}$$

$$\# \hat{p} \pm z_{\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$0 < \hat{p} < 1$$

$$\# E = z_{\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Sample size for estimation for p ?

$$n = \left(\frac{z_{\frac{\alpha}{2}}}{E} \right)^2 \hat{p}(1-\hat{p})$$

Application

Ex: A sample of 500 nursing application include 60 form men

Find 90% C.I of the proportion of men who applied to the nursing program.

Soln: $\hat{p} = \frac{x}{n} = \frac{60}{500} = 0.12$

$$\hat{p} \pm z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

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$$0.12 \pm (1.64) \sqrt{\frac{(0.12)(0.88)}{500}}$$

$$0.096 < p < 0.144$$

$$90\% \text{ C.I.} = 9.6\% , 14.4\%$$

H.w Among 1250 people we found 75 of them visited Petra. Find 99% C.I of the proportion of all people who visit petra?

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● Application

[43]

Ex 1 : A researcher wishes to estimate with 95% confidence. The proportion of people who own a home computer. A previous study shows that 40% of those interviewed had a computer at home. The researcher wishes to be accurate within 2% of the true proportion.

Find the minimum sample size? $\Rightarrow n = ?$

Sol : $\hat{p} = 0.4$
 $1 - \hat{p} = 0.6$
 $E = 0.02$

$1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$
 $z_{\frac{\alpha}{2}} = 1.96$

$$\Rightarrow n = \left(\frac{z_{\frac{\alpha}{2}}}{E} \right)^2 \cdot \hat{p}(1 - \hat{p}) = \left(\frac{1.96}{0.02} \right)^2 (0.4)(0.6)$$

$$= 9604 \times 0.24 = 2304.96$$

بنظر ال size للأكبر
 $\boxed{= 2305}$

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Ex 2 A simple random sample of 400 items provides 100 yes responses.



What is the point estimate of the 44 proportion of the population who would provide yes responses.

Find the standard error of the estimate.

Compute 90% C.I for the population proportion?

sol

a) $\hat{p} = \frac{x}{n} = \frac{100}{400} = \frac{1}{4} = 0.25$

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b) $S.E(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.25(1-0.25)}{400}} = 0.217$

c) $1-\alpha = 0.9$

$\Rightarrow \alpha = 0.1 \Rightarrow Z_{\frac{\alpha}{2}} = 1.64$

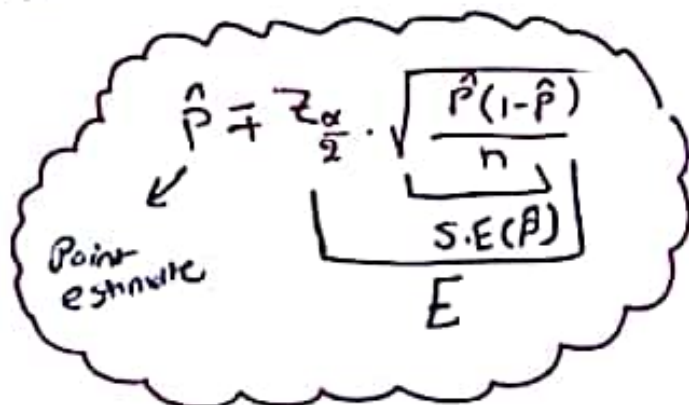
$\hat{p} \pm Z_{\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

$\Rightarrow 0.25 \pm (1.64) \sqrt{\frac{(0.25)(0.75)}{400}}$

$\Rightarrow 0.25 \pm 0.0356$

$0.2144 < \hat{p} < 0.2856$

$\hat{p} \in (21.44\%, 28.56\%)$



Ex ③ A simple random sample of $n = 25$ items from a normal population with μ (mean) and σ^2 resulted in a sample of $\bar{x} = 20$ & a sample standard deviation $S = 6$

$\bar{x} \Rightarrow$

Q Give a point estimate of μ .

45

Q Find standard error of $\bar{x}$?

Q Compute the 95% CI for the population mean?

sol Q $\bar{x}$ point estimate for $\mu \Rightarrow \hat{\mu} = \bar{x} = 20$

Q $S.E(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{6}{\sqrt{25}} = \frac{6}{5} = 1.2$

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Q σ : unknown & $n < 30$ (ni small)

$$\bar{x} \pm \underbrace{t_{\alpha/2, n-1}}_E \cdot \underbrace{\frac{s}{\sqrt{n}}}_{S.E(\bar{x})}$$

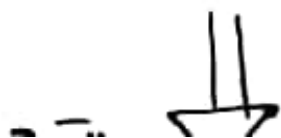
$$1 - \alpha = .95 \\ \alpha = 0.05 \\ t_{0.025, 24} = 2.064$$

$$20 \pm (2.064) * 1.2 \\ = 2.4768$$

$$17.5232 < \mu < 22.4768$$

$$\mu \in (17.5232, 22.4768)$$

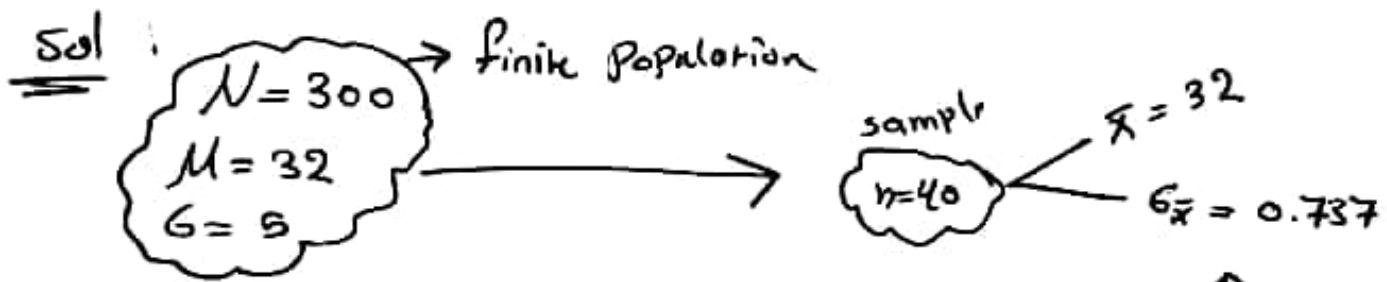
Ex (4) Assume that $\mu = 32$ & $\sigma = 5$ and the population has 300 items and that a simple random of 40 items is used the obtain information about this population



② what is the Expected value of $\bar{x}$?

③ " " standard deviation of $\bar{x}$?

④ what you can say about the standard deviation of $\bar{x}$ As the sample size increased?



② $E(\bar{x}) = \mu_{\bar{x}} = \boxed{32}$

③ $G_{\bar{x}} = S.D(\bar{x}) = S.E(\bar{x}) = \frac{G}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}$

$= \frac{5}{\sqrt{40}} \cdot \sqrt{\frac{300-40}{300-1}} = 0.737$

④ $S.D(\bar{x}) = G_{\bar{x}}$ decrease As a sample size increase

علاقة عكسية

Ex: you are a travel agent and you wish to estimate with 95% C.I the proportion of person who plan to travel outside next ten months. your estimate must be accurate within $\frac{E}{\text{true proportion}}$ of the true proportion.

Find the minimum sample size needed study found 18% person level outside Jordan?

sol: $n = \left(\frac{Z_{\frac{\alpha}{2}}}{E} \right)^2 \cdot \hat{p}(1-\hat{p})$

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$\hat{p} = 0.18$
 $E = 0.03$
 $Z_{\frac{\alpha}{2}} = 1.96$

$\Rightarrow n = \left(\frac{1.96}{0.03} \right)^2 \cdot (0.18)(1-0.18) = 466.67$
 $\leftarrow \boxed{= 467}$ بنقر

H.W Application ch(4)

Q Given the following D.O.F

$f(x) = \begin{cases} c & : 2 < x < 4 \\ 0 & : \text{o.w (other wise)} \end{cases}$

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① Find the value of c ?

② Find $P(x > 1)$?

③ Find $E(x)$ and $\text{Var}(x)$?

$\Rightarrow$ تمام الحالت

Ex: ch "S"

A population consists of numbers $\{1, 2, 3, 4\}$ In selecting random sample of size $n=2$ without replacement from this population:

① compute population mean μ (M)

Sol $\mu = \sum_{i=1}^4 \frac{x_i}{4} = \frac{10}{4} = \frac{5}{2} = 2.5$

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② Compute population variance (σ^2)

Sol $\sigma^2 = \sum_{i=1}^4 \frac{(x_i - \mu)^2}{N}$

نوعه نمونه
نمونه $\bar{x} \in M$ به صورت
نمونه $n-1$ از N

$= \frac{(1-2.5)^2 + (2-2.5)^2 + (3-2.5)^2 + 4-(2.5)^2}{4} = \boxed{1.25}$

③ Draw all possible samples of size $n=2$ & Find sampling distribution for the sample mean $\bar{x}$? pmf

Sol # of samples $= \binom{N}{n} = \binom{4}{2} = \frac{4!}{2!2!} = \underline{6}$



$$N = \{1, 2, 3, 4\}$$

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$$\{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}$$

$$X = \begin{matrix} \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 1.5 & 2 & 2.5 & 2.5 & 3 & 3.5 \end{matrix}$$

$\bar{x}$	frequency	prob
1.5	1	1/6
2	1	1/6
2.5	2	2/6
3	1	1/6
3.5	1	1/6

$\bar{x}$	1.5	2	2.5	3	3.5
$P(\bar{x}=x)$	1/6	1/6	2/6	1/6	1/6

P.m.f ($\bar{x}$)

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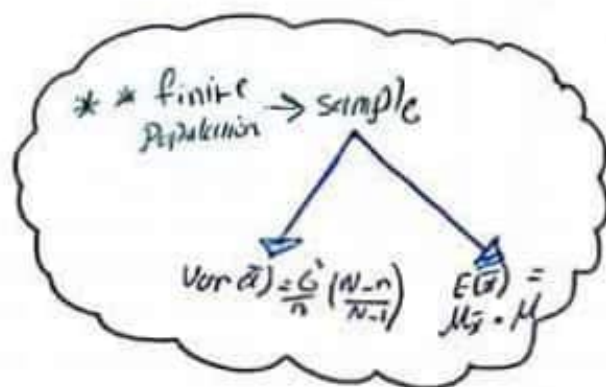
4) Find the mean (Expected value) of the sample mean $\bar{x}$? $E(\bar{x})$

5) Find $\text{var}(\bar{x})$?

6) Find S.D ($\bar{x}$)?

Sol 4) $\Rightarrow E(\bar{x}) = \mu_{\bar{x}} = \mu = 2.5$

or $E(\bar{x}) = \sum \bar{x} \cdot P(x = \bar{x})$



5) $\text{Var}(\bar{x}) = \frac{\sigma^2}{n} = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right) = \frac{1.25}{2} \left(\frac{4-2}{4-1} \right) = 0.1416$

6) $\sqrt{0.1416}$

ch (4)

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Assume that the test scores from a college admissions test are normally ~~distributed~~ with a mean of 450 & standard deviation of 100

① what percentage of the people taking the test score between 400 and 500?

② what that someone receives a score of 360 what percentage of the people taking the test score better? what percentage score worse?

sol $X: N(\overset{450}{\underset{M}{\cancel{400}}}, \overset{(100)^2}{\underset{6^2}{\cancel{100}}})$

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① $P(400 \leq X \leq 500)$ $Z: N(0,1)$

$$P\left(\frac{400 - 450}{100} \leq \frac{X - \overset{M}{\cancel{450}}}{6} \leq \frac{500 - 450}{100}\right)$$

$$P(-0.5 < Z < 0.5) = P(Z < 0.5) - P(Z < -0.5)$$

$$= \underline{0.303}$$

$\Rightarrow$ percentage 30.3%

$$\begin{aligned}
 \textcircled{6} P(X > 360) &= P\left(\frac{X - \mu}{\sigma} \geq \frac{360 - 450}{100}\right) \quad |52| \\
 &= P(Z \geq -0.9) = 1 - P(Z \leq -0.9) \\
 &= \boxed{91.99}(\%) \\
 &= 81.99\%
 \end{aligned}$$

better ~~worse~~ $\Rightarrow$ 81.99%

worse $\Rightarrow$ 18.31%

Q: Find the Area Under the part of a normal curve that lies within?

- ($\mu - 6 < x < \mu + 6$)
- | | | | |
|---|----------------------|--------------|---------|
| 1 | 1 standard deviation | of its mean? | = 68% |
| 2 | " | " | = 95% |
| 3 | " | " | = 99.7% |

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